MIL-T-11663K
14 December 1987
SUPERSEDING
MIL-T-11663J

MILITARY SPECIFICATION

TRUCK, MATERIALS HANDLING, CONTAINER HOISTING: REAR HOIST, 21,000 TO 32,000 POUNDS GVW, 4x2, COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel or gasoline engine driven, four-wheel, two-rear-wheel drive materials handling trucks having gross vehicle weights (GVW) of 21,000 to 32,000 pounds, for normal operating conditions. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The vehicle shall be one of the types, classes and sizes shown in table I, as specified (see 6.2):

TABLE I. Vehicle classification.

Type	Nomenclature	Class	Size
I	Arm and chain lift	A	1
		B	2
		C	3
		D	4
II	Rigid fork arm, high lift	D	2

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-2320

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1.2.1 Class. The class of the vehicle shall be determined by the minimum gross vehicle weight (GVW) rating as specified in table II.

TABLE II. GVW ratings.

Vehicle class	GVW ratings (pounds)
A	21,000
B	24,000
C	28,000
D	32,000

1.2.2 Size. The size of the hoisting unit shall be determined by the minimum rated lifting capacity as specified in table III.

TABLE III. Size.

Size	Lifting capacity (pounds)
1	7,500
2	9,000
3	12,000
4	15,000

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage, Vehicular, Ignition, Lighting and Starting.
ZZ-T-381 - Tires, Pneumatic, Vehicular (Highway).

MILITARY

MIL-R-22827 - Refuse Containers, Hoisting Units.
* MIL-H-83282 - Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H-537.

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STANDARDS

FEDERAL

* FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

* MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.

* 2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

Federal Motor Vehicle Safety Standards.

* (Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government document which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM Standards
ASTM B117 - Method of Salt Spray (Fog) Testing.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SAE, INC.
SAE Standards and Recommended Practices.

	J537	- Storage Batteries.
*	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation From Vehicles and Devices (30-1000 MHz).
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
	J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
	J688	- Truck Ability Prediction Procedure (DoD adopted).
	J704	- Openings for Six and Eight-Bolt Truck Transmission Mounted Power Take-Offs.
	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicles and accessories. Except as specified in 3.1.1 through 3.1.1.5, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items which meet or exceed the requirements of this specification. All

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chassis items shall be as represented in the chassis manufacturer's technical data. Special bodies, mounted hoist-haul unit, and equipment shall be as represented and rated in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of the invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment, painting, marking and data plates. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color and the identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2).

* 3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle, except the truck body, shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.3 Drain plugs. Drain plugs installed in the manual transmission and rear axle(s) for draining lubricants and drain plugs installed throughout the vehicle for draining hydraulic fluids shall be of the permanent magnet type.

3.1.1.4 Towing devices. Not less than two hooks, loops, pins, or a single, center-mounted towing eye for towing the vehicle shall be furnished on the front of the vehicle.

* 3.1.1.5 Wheel splash and stone throw protection. The vehicle shall have rigid splash shields ahead of the rear wheels and rubber mud flaps to the rear of the rear wheels. All tilt cabs shall have rubber mud flaps to the rear of the front wheels. A metal strip not less than 1/8 inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. All splash shield and mud flap installations, front and rear, shall conform to SAE J682.

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3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior drive-by sound level shall conform to the EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories and equipment; the body; the hoist-haul unit; and a full complement of fuel, lubricants, coolant and hydraulic fluid.

3.2.5 Payload. The payload shall consist of the total weight of the container and its contents and shall be equal to the rated lifting capacity of the specified size of the hoist-haul unit. The payload distribution between the front and rear axles shall be based upon the sum of the weight of the container specified in table IV and its contents resting in the travel position of the hoist-haul unit.

TABLE IV. Container lift size.

Hoist unit size	Container size (cubic yards)
1	2.5
2	3.0
3	4.0
4	5.0

3.2.6 Gross vehicle weight. The gross vehicle weight (GVW) shall consist of the curb weights, the payload in travel position, and the operator weight (computed at 175 pounds). The rated capacities of the suspension, axles and tires shall be adequate to meet the minimum GVW and in addition shall meet the imposed weight when loaded as specified in 3.2.5.

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3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicle ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be available to the engineering office of the procuring activity.

3.2.7.1 Hoisting unit. The rated lifting capacity of the hoisting unit shall be based on the hoisting unit being capable of lifting the rated load specified in 1.2.2 when handling a container conforming to MIL-R-22827 of the size specified in table IV.

3.2.8 Dimensions. Cab-to-axle and back of cab to back of rear spring rear hanger bracket dimensions shall be as recommended by the manufacturer of the body hoisting unit. The overall width of the vehicle, exclusive of mirrors, lights, reflectors and tires, shall be not more than 96 inches. The width over the tires shall be not more than 100 inches.

* 3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the vehicle loaded as specified herein (see 3.2.5 and 3.2.6).

3.3.1.1 High speed gradeability. The vehicle shall ascend a continuous 1.0 percent grade at 50 miles per hour (mph). Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3). Gradeability requirements shall be met with the main transmission in direct drive, and when a multispeed axle or auxiliary transmission is furnished, with the axle or auxiliary transmission in high speed range.

3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed and shall provide a vehicle speed not greater than 4 mph.

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to the geared speed specified shall be determined by the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3)}}$$

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3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52, under all conditions of loading, up to the specified GVW.

3.4 Chassis components.

* 3.4.1 Engine. The engine furnished for the specified vehicle class shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine with not less than four cylinders on class A and class B trucks and not less than six cylinders on class C and class D trucks. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J1349.

* 3.4.1.2 Gasoline. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at an engine speed of not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel, when used in accordance with the operator's manual. Net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J1349. Unless otherwise specified or required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline.

3.4.1.3 Oil filter. A full flow type oil filter shall be furnished.

3.4.1.4 Cooling system. The cooling system shall maintain engine coolant at a temperature below the boiling point with the vehicle loaded to the rated GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity. A high coolant temperature or low coolant level alarm buzzer and a red indicator warning light shall be provided on the dash instrument panel. On tilt cab models, a radiator servicing access door shall be provided to allow verification of the coolant level.

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3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters or complete thermostatic control of all coolant flow through the radiator.

3.4.1.6 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically, when the fan is not required for engine cooling.

3.4.1.8 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

3.4.1.9 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater and a fuel warmer (diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. The heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block to a single three-pronged (male), weatherproof slave receptacle for receiving external power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). The electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

* (a) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. An engine coolant thermostat with an operating range of 170° to195°F shall be installed
(b) Immersion type engine oil heater, 300-watt minimum rating with 170° to 195°F thermostat, shall be installed in the oil pan through any convenient opening

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(c) Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as minus 60°F, and shall embody a thermostat to limit the temperature of electrolyte to not more than plus 80°F

(d) A fuel warmer or pre-heater shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

* 3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12or 24-volt direct current (dc) starting system, with 12-volt dc lighting system shall be furnished. Unless otherwise specified (see 6.2), a minimum 60-ampere alternator shall be furnished. Engine starting equipment shall include an ether starting system or glow plug. If an other system is furnished in lieu of a glow plug, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment, and shall be inoperative with the engine warm. Complete provisions for a replacement ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished. Unless otherwise specified (see 6.2), an alternator of not less than a 40-ampere rated capacity which provides not less than 16 amperes dc output at normal engine idle speed shall be furnished.

* 3.4.2.3 Lighting. All vehicle lights and reflectors shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20, and 393.22 through 393.26(d). When available as the manufacturer's standard or optional feature, the brake lights shall override the four-way emergency flasher. Lights and reflectors shall not be mounted on vertical surface of rub rails (unless recessed and fully protected) or mounted on vehicle bumpers.

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* 3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. The vehicle shall be provided with double-faced front signal units and with single-faced rear signal units installed in accordance with SAE J588, except that turn signals shall not be mounted on the engine compartment hood. Turn signal operating units shall have visible and audible type flash indicators.

* 3.4.2.5 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity ratings and the total cold cranking ampere ratings at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table V. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

TABLE V. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	320	1,200
Gasoline	100	450

3.4.2.6 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tank(s). Fuel tank(s) shall be of not less than 50 gallons total capacity (43 gallons for vehicles destined for California). When more than one fuel tank is furnished on diesel engine driven vehicles, means shall be provided to assure equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve connecting either tank to the engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage.

3.4.3.3 Fuel and water separator. When specified (see 6.2), in addition to the manufacturer's standard fuel filter(s), a fuel and water separator shall be furnished. The separator shall include a water coalescer and a drain valve.

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* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Regulation 393.83. Vertical exhaust systems, if furnished and if capable of being reached easily by personnel entering or leaving either side of the cab, shall be provided with a heat shield. Vertical exhaust systems shall be provided with a hinged rain cap.

3.4.4.1 Exhaust system safety shield. The exhaust system or the hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

3.4.5 Transmission. Unless otherwise specified (see 3.4.5.2), a manually shifted transmission with not less than five forward speeds shall be provided. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. When more than five forward speeds are required, a multispeed transmission or auxiliary transmission may be provided. Gear ratios in the transmission and axle shall be matched to provide a progressive shifting pattern throughout the complete range and shall provide the required vehicle performance (see 3.3.1 through 3.3.1.3). The transmission shall be provided with SAE J704 power takeoff openings at the right and left sides. The transmission shall provide for maximum ease of shifting in all speeds.

* 3.4.5.1 Clutch. The clutch shall be the largest capacity clutch offered for the type and class vehicle and engine furnished, with the clutch torque capacity exceeding the maximum delivered engine torque.

* 3.4.5.2 Automatic or semiautomatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic or semiautomatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios. The normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with SAE J704 power takeoff opening(s).

3.4.5.3 Power takeoff. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT DRIVE VEHICLE WITH POWER TAKEOFF ENGAGED" shall be installed so as to be readily visible to the seated driver.

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3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine as developed through the maximum gear train reduction.

3.4.7 Frame. The vehicle shall be provided with a continuously flat mounting surface from the rear of the cab to the extreme end of the frame rails. Type I vehicles shall have a frame with reinforcement extending at least from the rear of the front suspension, rear hanger bracket to the front of the rear spring, front hanger bracket. Type II vehicles shall have a frame with a reinforcement extending the full length, front to rear. The reinforcement shall provide sufficient structural strength in the chassis frame, through increased resisting bending moment, to at least equal the loads imposed with the vehicle loaded to the maximum capacity specified.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system, with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded as specified in 3.2.5 and 3.2.6. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded as specified in 3.2.5 and 3.2.6. When specified (see 6.2), the wheel bearings and axle spindles shall be oil lubricated. The hubcaps shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure build-up and for replenishing the oil supply shall be provided.

* 3.4.9.1 Two-speed axle. When specified (see 6.2), a two-speed shift axle shall be furnished, equipped with an electric, vacuum, or air shift and provided with ratios which will permit proper gear ratio selection to meet the vehicle performance specified (see 3.3.1 through 3.3.1.3). Vehicles provided with an automatic or semiautomatic transmission shall be equipped with a single speed axle.

3.4.9.2 Traction control. The vehicle shall be furnished with a traction control on the rear axle. The traction control shall actuate automatically to ensure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

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3.4.10 Wheels, rims, tires, and tubes. Unless wide base tires are specified, the vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Rims and disc wheel sizes shall be the same for all wheels on each vehicle. Tire size and load range (ply rating) shall be the same for all tires on each vehicle. When specified (see 6.2), the vehicle shall be equipped with wide base type tires and wheels for the front and rear axles in lieu of conventional front and dual rear wheels and tires. Wide base wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), 10-stud disc type wheels shall be furnished.

3.4.10.1 Tires. Steel belted radial or, when specified (see 6.2), bias ply tires shall be furnished. Tires shall be tube or tubeless type with highway tread. Tires shall be of a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded as specified in 3.2.5 and 3.2.6. Tires shall conform to Tire and Rim Association recommendations, or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association,

3.4.10.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of the heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.4.11 Brakes. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52. When available as the manufacturer's standard or as an option, brake linings shall be of nonasbestos material.

* 3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7-1/4 cubic feet per minute (cfm)
* (b) Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
(i) Automatic slack adjusters.

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* 3.4.11.2 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.3 Brake controls for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, a relay emergency valve with no-bleed-back feature (except when a spring applied emergency brake is furnished) and additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. Unless otherwise specified, the chassis manufacturer may furnish any type of his standard or optional full-width cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanisms, shall be furnished. When specified (see 6.2), a tilt type cab with a tilting and locking mechanism shall be furnished. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 24 inches. The cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with at least the curbside door equipped with an external, key-operated lock. Drip rails shall be installed above the cab doors. Unless otherwise specified, the cab shall have an upholstered, full-width, adjustable seat and back or individual, adjustable driver's seat and individual passenger seat. When specified (see 6.2), an individual, adjustable driver's seat and an individual passenger seat shall be provided. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished interior lighting shall be provided. Two pairs of seat belts shall be installed. Safety grips or grab handles shall be provided on each side of the cab to assist personnel in entering and leaving the cab. For tilt type cabs, provisions to facilitate cleaning the windshield shall be provided by means of a bumper step, or bumper step cutouts, and a grab handle located under the windshield.

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3.4.12.1 Periscope. When specified for type II (see 6.2), a periscope type mirror screen shall be installed in the ceiling of the cab to enable the operator to view the container during all operating positions.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be of the multispeed type operated by either air or electric motor(s).

3.4.15 Bumper. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided on type I vehicles. On type II vehicles, front counterweights, when used, shall serve as a front bumper (see 3.5.2.3).

* 3.4.16 Tool stowage. Stowage space of sufficient size to accommodate hand tools, antiskid chains and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools and equipment may be furnished in the cab. When stowage space for these tools is located outside of the cab, it shall be weatherproof and shall provide for locking with a padlock or a key operated lock.

3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to the floor and to windshield defroster louvers. The heater shall be complete with a blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be panel mounted and visible to the driver when seated in a driving position.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

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(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or indicator
(e) Engine temperature gage or indicator
(f) Speedometer with recording odometer
(g) Ash receptacle
(h) Dual sunvisors
(i) Front door mounted armrest on driver's side
(J) Driver's compartment ventilator other than window
* (k) Tachometer.

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 20 square inches of reflective area. The mirrors shall have not less than two supporting arms.

3.4.21 Horn. Manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be provided.

* 3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to ensure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.23 Back-up alarm. The vehicle shall be provided with an audible, pulsating, signaling device (electrical or mechanical) to caution personnel when the vehicle is in reverse gear operation.

3.5 Hoist-haul unit. A hydraulically operated hoist-haul unit shall be furnished and shall be capable of handling all the types and sizes of containers specified in MIL-R-22827. The operation of the hoist-haul unit shall be controlled from the cab. The hoist-haul unit shall be capable of accomplishing its intended purpose without modification of either the hoisting unit or container. The unit shall be complete with rear fenders, cab controls, and all necessary items for mounting. An air or hydraulically operated retaining device for each upper and lower dumping hook assembly shall be provided. The device shall be furnished with cab mounted controls,

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designed for positive retention of the container bail when engaged in the dumping hook, and shall be provided to prevent the possible disengagement of the container from the carriage during dumping operations. The dumping hook shall automatically engage the container bail. A spring shall be provided and designed to exert 60 pounds of force on the dumping hook to assure a receptive position for the container bail engagement. Engagement and disengagement of the dumping hook retention devices shall be controlled by the operator from the cab. An indicator light shall be furnished in the cab and marked to indicate to the driver that the dumping hook container bail retaining devices are engaged in the locked position. An instruction decal shall be provided in the cab to explain the complete sequence of the operation. The dumping hooks on type I units shall be automatically retracted and retained by the forward horizontal movement of the carriage assembly to prevent inadvertent latching of the container bail so as to allow proper placement of the container in the travel position. The dumping hooks shall be exposed for the container bail engagement only when the carriage assembly is in the rearmost position.

3.5.1 Type I. Type I unit shall be the chain hook-up and two-arm lift type, for lifting, placing on the truck, transporting to a dump area, and dumping the contents of the container. The hoist-haul unit shall be designed and mounted on the truck chassis to operate and perform satisfactorily without truck counterweights.

3.5.1.1 Performance. The unit shall be capable of picking up any container from ground level and from 45 inches above ground. After attachment, each of the following fully loaded container pickup operations shall be accomplished in not more than 1 minute:

(a) From ground to travel position
(b) From 45 inches above ground to travel position
(c) From travel position to dumping contents and back to travel position.

3.5.1.2 Dumping. Any fully loaded container shall be completely emptied when contents are dumped in a pit or trench located not less than 4 feet rearward of the rear axle.

3.5.1.3 Travel position. A continuous support for the bottom of all containers shall be provided on the truck to carry the containers while the vehicle is in transit. When the container is in the carrying position, no part of the weight of the container, with payload, shall be supported by the lifting arm, lifting chains and hydraulic mechanism. Provision shall be made for positively retaining containers in the carrying position to prevent movement while in transit.

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3.5.1.4 Body. The body shall consist of a subframe, twin boom assembly, carriage, skid frame, heavy duty rear apron and fenders over the rear wheels. The booms shall be parallel to each other and locked together to act as a unit. The booms shall be actuated by hydraulic cylinders. The end of each boom shall be furnished with a lifting chain and means for attaching to pins on the containers. The attachments shall be designed to prevent the container from being accidentally detached. The skid frame shall be permanently attached to provide support to the bottom of the container in the carrying position. The carriage shall provide support for the back of the container and shall guide the container into the carrying position.

3.5.1.5 Stabilizing jacks. Rear mounted stabilizing jacks, capable of being operated from the cab, shall be furnished for class C and D vehicles. The jacks shall provide adequate stability with relation to the ground while operating the hoist-haul unit under maximum loads. A red indicator light, to warn the driver that the jacks are down, shall be installed in a conspicuous location in the cab.

3.5.2 Type II unit. Type II unit shall consist of a three-stage telescopic mast with two controlled rear fork lifting arms slipping under the container lifting pins. The unit shall be designed for use where high lifting of containers for dumping refuse is required. The unit shall provide for one-man operation, with container pickup not requiring the operator to leave the vehicle cab.

3.5.2.1 Performance. The carriage shall be capable of tilting through an arc of approximately 30 degrees (25 degrees forward and 5 degrees rearward of the vertical). When the carriage is fully elevated, the height from the ground to the top of the forks, when fully loaded, shall be approximately 13 feet. The time required for the complete dumping cycle from travel position and return to travel position, shall be not more than 2 minutes.

3.5.2.2 Body (hoist-haul unit). The unit shall consist of a body frame; hydraulically operated lifting fork arms with flanged guide tips and container-retaining notches; air-operated upper and lower dumping hooks; and fenders provided over the rear wheels. The fork sections shall be provided with gusset reinforcements.

3.5.2.3 Counterweights. Counterweights shall be provided and installed on the front of the truck chassis or, when the front bumper is an integral part of the cab, behind the cab. Counterweights shall provide proper balance for the fully elevated and loaded container in vertical position on the carrier. Front counterweights, when used, shall also serve as a front bumper.

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3.5.3 Hydraulic systems. The hoist-haul units shall be hydraulically operated. Power shall be transmitted from a transmission mounted power takeoff through a heavy duty gear or vane type pump. The normal operating pressure range shall be between 900 pounds per square inch (psi) and 1600 psi. Relief valves shall be incorporated into the hydraulic system to prevent excessive pressure. A line tap shall be provided so that a pressure gage may be installed to periodically check relief pressure. All hydraulic lines, hoses, and fittings shall meet or exceed the requirements of SAE recommended practices and standards. The size, volumetric capacity, and pressure rating of all components shall be such as to assure a smooth and efficient operation of the system.

* 3.5.3.1 Hydraulic fluid. Unless otherwise specified, the manufacturer's standard or recommended hydraulic fluid shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system. A permanent decal located near the hydraulic system fill port shall be furnished stating "CAUTION: USE HYDRAULIC FLUID CONFORMING TO MIL-H-83282 ONLY."

* 3.5.3.2 Hydraulic cylinders. All hydraulic cylinders shall be fitted with seals to prevent entrance of dirt, moisture, and other foreign elements and to preclude leakage of hydraulic fluid from the cylinders. Hydraulic piston rods shall be chrome-plated full length with sufficient thickness to withstand a 30-hour salt spray test in accordance with ASTM Designation B117. The connecting ends shall be treated in accordance with the best commercial practice.

3.5.3.2.1 Type I cylinders. Two vertical cylinders shall be provided to control the elevating and lowering of the boom arms. One horizontal cylinder shall be provided to control the forward and rearward movement of the carriage for container pickup, moving the container into travel position, and moving the container into position for dumping refuse.

3.5.3.2.2 Type II cylinders. One single-acting cylinder shall be provided to control lifting and lowering of the forks, two double-acting cylinders to control the dumping action, and one double-acting cylinder to control the tilting action of the carriage.

3.5.4 Lubrication. All moving parts and surfaces of the equipment normally requiring lubrication shall be provided with lubrication fittings of the hydraulic type with surface check ball and, when applied to sealed bearings, shall have pressure release to protect the seals.

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3.5.5 Controls and operating mechanisms. All controls and operating mechanisms shall be located inside the truck cab. Controls shall be complete and conveniently operable by driver. All operations (lifting, dumping, lowering, and tilting) shall be controlled by the operator from his normal position in the cab, without outside assistance. Controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments inside the cab shall be visible to the driver when seated in the normal driving position.

* 3.5.5.1 Controls and safety. The hoisting mechanism controls shall be located in the cab of the truck. The position of the container and hoist arms shall be under the control of the operator at all times, so that the operator can start, stop or reverse the movement at any point in the cycle. The hoisting system shall be designed to minimize the possibility of injury to operating personnel from the hoist arms or container, and shall be designed to allow safe entrance and exit to and from the cab by the operator and passenger(s) during any phase of the hoisting and dumping operations.

3.5.6 Mounting. The hoist-haul unit shall be mounted on the truck chassis in accordance with the current highest standard practice of the manufacturer of the hoist-haul unit.

* 3.6 Servicing and adjusting. Prior to acceptance of the vehicles by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following; alinement of lights; adjustment of the engine and brake systems; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete lubrication of the chassis, hoisting unit, engine and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted and riveted construction utilized shall be in accordance with the highest standards of the industry.

3.7.1 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the

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parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subject to proof and service loadings. Fillet welds shall be provided when necessary to reduce stress concentration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of the product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle of each type and class (see 1.2) produced under the contract shall be subjected to an inspection and a road test. The inspection and road test shall be conducted by the contractor, at or near his plant, under the direction and in the presence of Government representatives. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the contract.

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4.3.1 Vehicle weight. To determine conformance to 3.2.4 through 3.2.7, the first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axle. The total imposed loading on front and rear axle shall be computed by the contractor and verified by the Government, using the curb weight, the operator weight at 175 pounds, and the payload (see 3.2.5) to provide the required GVW. Calculated imposed loads on front and rear axles shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

4.3.2 Road test. To determine conformance to 3.3 through 3.3.2, the first production vehicle, with rated payload, shall be road tested for a distance of not less than 20 miles at speeds up to 50 mph. At least 20 percent of the road test distance shall be over dry, unimproved or gravel roads at speeds up to 20 mph. The vehicle shall be subjected to not less than three sudden stops from speeds of not less than 20 mph on a substantially level, dry, smooth surface free of loose material. The vehicle shall be observed for the specified stopping distance and for exceeding the deviation from the specified width lane. Upon completion of the road test and prior to hoist-haul unit test (type I and II), the vehicle, all equipment, components and accessories shall be examined for damage and leakage (lubricant, air or hydraulic).

4.3.2.1 Type I hoist-haul unit test. To determine conformance to 3.5.1 through 3.5.1.3, type I hoist-haul unit (chain hook-up and two-arm lift type) shall be subjected to a performance test. An applicable type container conforming to MIL-R-22827 shall be furnished for the test. The container pick-up and dumping operations shall be performed without counterweights. The time period of not more than one minute to accomplish the operations shall be verified. During the test, the following verifications and inspections shall be performed:

(a) After attachment of the container and prior to pickup, assure that the indicator light in the cab does indicate to the driver that the dumping hook container bail retaining devices are engaged in locked position
(b) Container pickup from ground to travel position. With the container in travel position, observe that no part of the weight of the container is supported by lifting arms, lifting chains or hydraulic mechanism. Also assure that provisions have been made for positively retaining containers in the carrying position to prevent movement during transit
(c) Verify container pickup from 45 inches above the ground to the travel position

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(d) Movement of container from travel position to dumping contents and back to travel position shall be observed. After dumping operation, observe that the container is completely emptied when contents are dumped in a pit, or trench located not less than four feet rearward from the rear axle.

4.3.2.2 Stabilizing jacks test. To determine conformance to 3.5.1.5, the rear mounted stabilizing jacks for class C and D vehicles shall be tested. During operation of the jacks system from the cab, the jacks shall be observed for proper operation without binding or malfunction. When the jacks are completely down, verify that the red indicator light is in operation to warn the driver that the jacks are down. With the jacks in the down position, the vehicle and hoist-haul unit shall be observed to verify that the jacks provide adequate stability with relation to the ground, while the hoist-haul unit is operating under maximum load.

4.3.2.3 Type II hoist-haul unit test. To determine conformance to 3.5.2 through 3.5.2.3, the type II hoist-haul unit (rigid fork arm high lift) shall be subjected to a performance test. The container pick up (in maximum elevated position) and dumping operations shall be performed with vehicle counterweights attached. The required time of not more than 2 minutes for the complete dumping cycle, from travel position and return to travel position, shall be verified. During the test, the following verifications and inspections shall be performed;

(a) When the carrier is in the maximum elevated position, with a fully loaded container, the approximate 13 feet height from the ground to the top of the forks shall be measured
(b) The carrier capability of tilting through an arc of approximately 30 degrees (25 degrees forward and 5 degrees rearward of the vertical) shall be verified
(c) After the dumping operation, observe that the container has completely emptied
(d) When the fully elevated and loaded container is in vertical position on the carrier, the vehicle and hoist-haul unit with container shall be observed to verify that the counterweights provide proper balance.

* 4.3.3 Truck body treatment and painting. The certification regarding the body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be made to Government representatives for examination and approval.

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4.3.4 Examination. Upon completion of testing the hoist-haul unit, the first production vehicle of each type shall be subjected to examination. The vehicle, with all equipment, components, accessories and container furnished for the test shall be examined for damage. The applicable hoist-haul unit shall be examined for damage, leakage (lubricant, hydraulic or air), evidence of distortion, permanent deformation or other malfunctions.

4.3.5 Production sample. Upon acceptance of the first production vehicle of each type and class, it shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

* 4.3.6 Failure. Failure of the first production vehicle(s) to meet the requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

4.4 Production vehicle inspection. Each production vehicle shall be subjected to dimensional inspection and operational tests (less payload) as specified in section 3. The vehicle, hoist-haul unit and associated controls shall be operated to the extent necessary to verify compliance with contractual requirements. The contractor's inspection system, as a minimum, shall assure that the vehicle and attached equipment conforms to the dimensional and operational requirements and the vehicle is capable of meeting the requirements herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 have been accomplished.

* 4.5 Contractor's records. The contractor's inspection records as required by the contract shall show the results of the first production vehicle inspection and tests and subsequent production vehicle inspections and tests. The records shall show the actual findings and any corrective action taken. The records shall be complete, traceable to the applicable vehicle and available to the Government. Inspection and test records shall be signed and dated by a person specifically assigned by the contractor, after satisfactory completion of each inspection or test. The Government will countersign this record after satisfactory completion of the first production vehicle inspection.

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5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The trucks are intended for general nontactical use at military installations for picking up and depositing containers, carrying containers to and from the refuse discharge area, and dumping the contents of the containers. In addition, type II trucks are intended for dumping where containers have to be lifted to high levels, such as dumping into trucks and semitrailers.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and size of vehicle required (see 1.2).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).
(d) Rustproofing, if required (see 3.1.1.2).
(e) Tropical rustproofing, if required (see 3.1.1.2).
(f) A gasoline engine, if required in lieu of a diesel engine (see 3.4.1.2).
(g) Silicone rubber hoses, if required (see 3.4.1.8).
(h) Power plant heaters, if required (see 3.4.1.9).
(i) If a battery heater is not required with power plant heaters (see 3.4.1.9).
(j) Alternator capacity, if other than as specified (see 3.4.2.1 and 3.4.2.2).
(k) A fuel and water separator, if required (see 3.4.3.3).
(1) Automatic or semiautomatic transmission, if required (see 3.4.5.2).
* (m) Oil lubricated wheel bearings and axle spindles, if required (see 3.4.9).
(n) Two-speed axle, if required (see 3.4.9.1).
(o) Wide base type tires and wheels, if required (see 3.4.10).
(p) Disc type wheels, if required (see 3.4.10).
(q) Bias ply tires, if required (see 3.4.10.1).
(r) Brake controls for use from a towing vehicle, if required (see 3.4.11.3).
* (s) Tilting hood and fender assembly, if required (see 3.4.12).

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(t) Tilt type cab, if required (see 3.4.12)
(u) Individual driver and passenger seats, if required (see 3.4.12).
(v) Periscope for type II vehicles, if required (see 3.4.12.1).
(w) Air operated horn, in addition, if required (see 3.4.21).

6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet item 1 should include vehicle model number, engine model number and vehicle type and class. Unless other conditions are cited in the contract, computations should be for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting truck ability (see 3.3.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. .208
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Fctor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, 1000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0

* 6.4 Subject term (key word) listing.
Container hoisting
Lifting capacity
Stabilizing jack
Telescopic mast
Truck, commercial
Truck, nontactical
Truck, materials handling, container hoisting.

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6.5 Identification of changes. The margins of this document are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

Custodians:
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Preparing activity:
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Project No. 2320-0334

Review activity:
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